

## **Ismail Mujtaba Khan**

**Product / Mechanical Engineer** — Automated Manufacturing & Hardware Prototyping

Contact: 302-444-2946

Ismailmujtaba77@yahoo.com

### **Education**

#### **University of Delaware**

M.S in Robotics & Automation Engineering.

Graduation Date: May 2025

Relevant Coursework: Advanced Topic: Computer Applications, Linear Systems, Intro to Robotics, Artificial Intelligence, Independent Study.

#### **Osmania University**

Hyderabad, India

B.E in Mechanical Engineering

Graduation Date: 2023

---

### **Skills**

**Mechanical & Manufacturing:** Mechanical Design, Prototyping, DFM, Tooling, Composite Manufacturing, Additive Manufacturing.

**CAD & Simulation:** Onshape, SolidWorks, AutoCAD, CREO, CATIA.

**Robotics & Automation:** Robotics Integration, Motor Control, Control Systems, ROS2 (basic)

**Software & Programming:** Python, MATLAB, MS Office Suite.

**Soft Skills:** Problem Solving, Critical Thinking, Public Speaking, Teamwork, Time Management, Communication.

---

### **Professional Experience**

#### **CarbonForm**

Newark, DE

Product/Mechanical Engineer

June 2025 – Present

- Played a central role in the design and development of the Advanced Automated Manufacturing Hardware System, translating concepts into functional prototypes through iterative mechanical design and testing.
- Conducted hands-on prototyping and design validation experiments, refining designs based on test results to improve reliability and manufacturability.

#### **Composite & Additive Manufacturing Lab, Center for Composite Materials, University of Delaware**

Newark, DE

Research Assistant

December 2023 – May 2025

- Co-authored research papers and contributed to a DOE (ARPA-E) funded project focused on energy-efficient composite manufacturing.
- Designed, fabricated, and tested advanced composite structures including drone frames, automotive B-pillars, bicycle frames, and robotic components, while supporting automation and path-planning efforts.

#### **Daikin Air-Conditioning**

Hyderabad, India

Intern

September 2022 – December 2022

- Contributed to manufacturing analysis and reporting activities, gaining exposure to production processes, quality tracking, and operational coordination.

**Maruti Suzuki India Limited (RKS Saboo Motors)**

Hyderabad, India

Intern Engineer

August 2022 – December 2022

- Supported engineers and technicians on the production floor, gaining hands-on exposure to manufacturing workflows, troubleshooting, and quality issues.

**CPDDAM Lab, Osmania University**

Hyderabad, India

Intern

August 2022 – October 2022

- Assisted with setup and execution of automated fabrication experiments, working hands-on with process iteration and testing.

---

**Leadership & Activities**

**IEEE PES (Power and Energy Society)**

Vice-Chair

May 2022 – May 2023

- Organized events and competitions and led multiple fundraising activities.

**Make A Difference (MAD), Nationwide NGO**

Foundational Skills Volunteer

January 2022 – April 2023

- Tutored underprivileged children, assisting them in excelling in their final exams.

---

**Projects**

- Automation of Continuous Carbon Fiber Composite Fabrication
- 3D Mapping of UD Lewes Campus using Aerial Drones
- Path Planning using April Tags and SLAM on Gazebo

---

**Organizations**

- Vice-Chair, IEEE PES (Power and Energy Society), 2022-2023
- Member, Horn Entrepreneurship Club
- Member, UD Archery Team
- Member, UD Fencing Team
- Member, UD Sailing Team
- Ambassador, IEEEEXTREME 16.0
- Volunteer, YOUVA Student-Based NGO, 2019-2023
- Member, SAE INDIA (Society of Automotive Engineers), 2019-2023